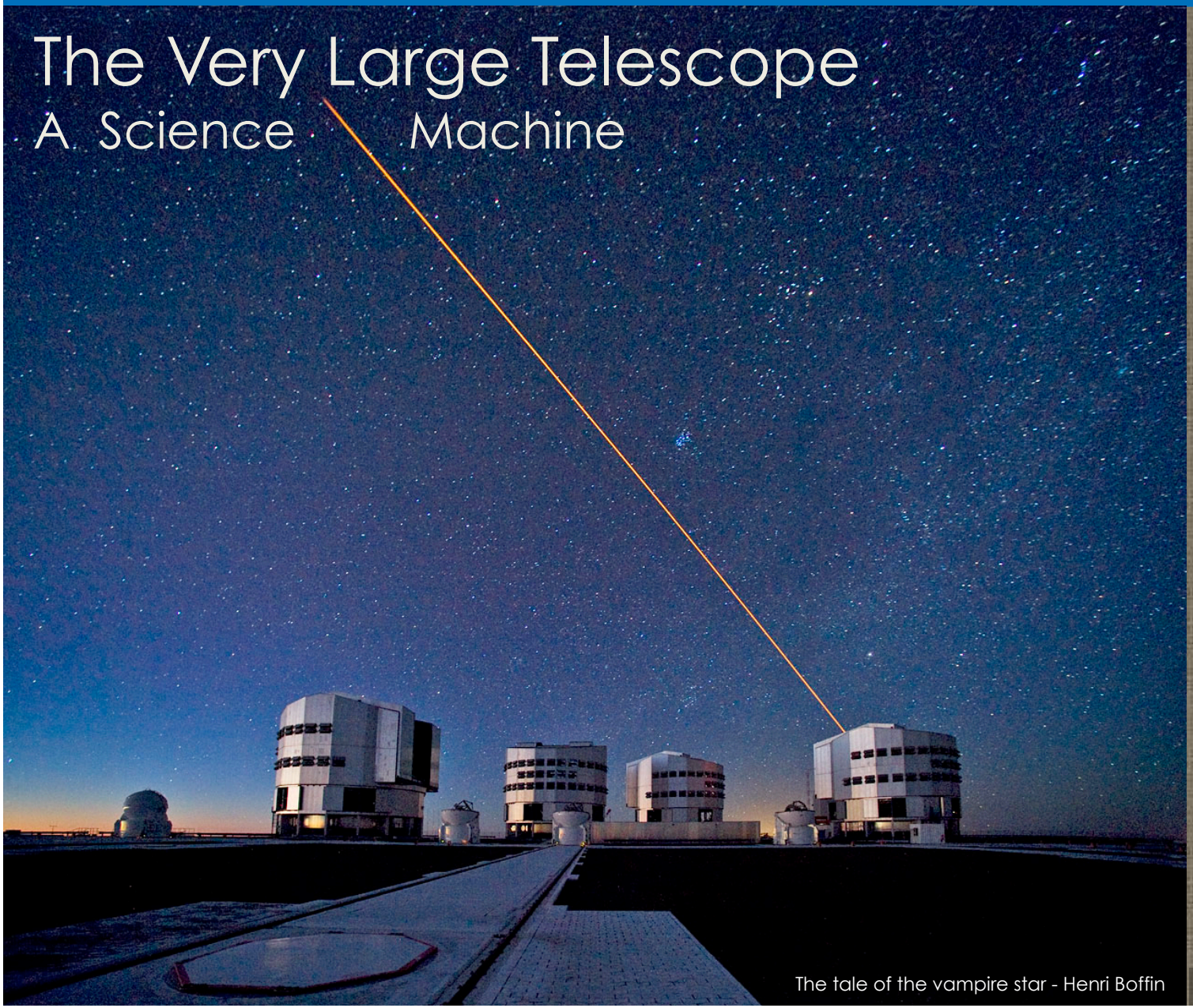
The Tale of the
Vampire Star
Henri M.J. Boffin
ESO

The Very Large Telescope
A Science Machine



Equipped with latest technology:

Active optics

Adaptive optics

Laser Guide Star

Interferometry

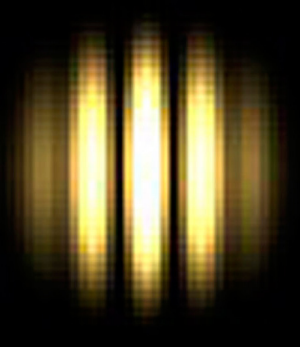
The tale of the vampire star - Henri Boffin

Interferometry: combining waves

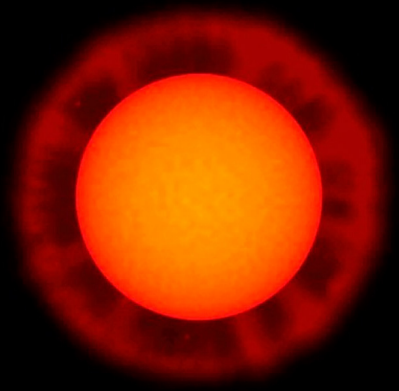
Measuring the radius of stars

Stars of different sizes

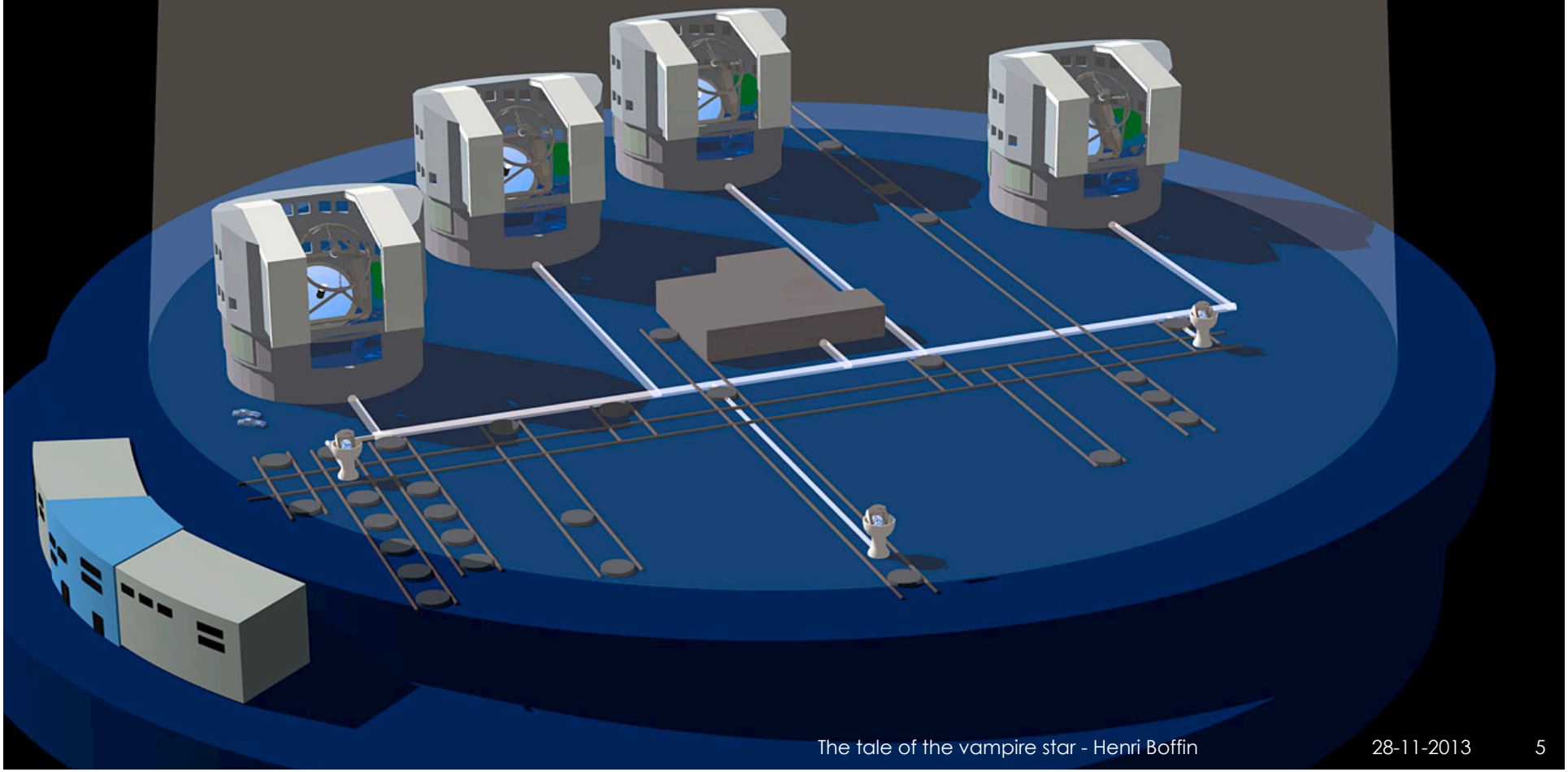
Seen through normal telescopes, it is not possible to distinguish them



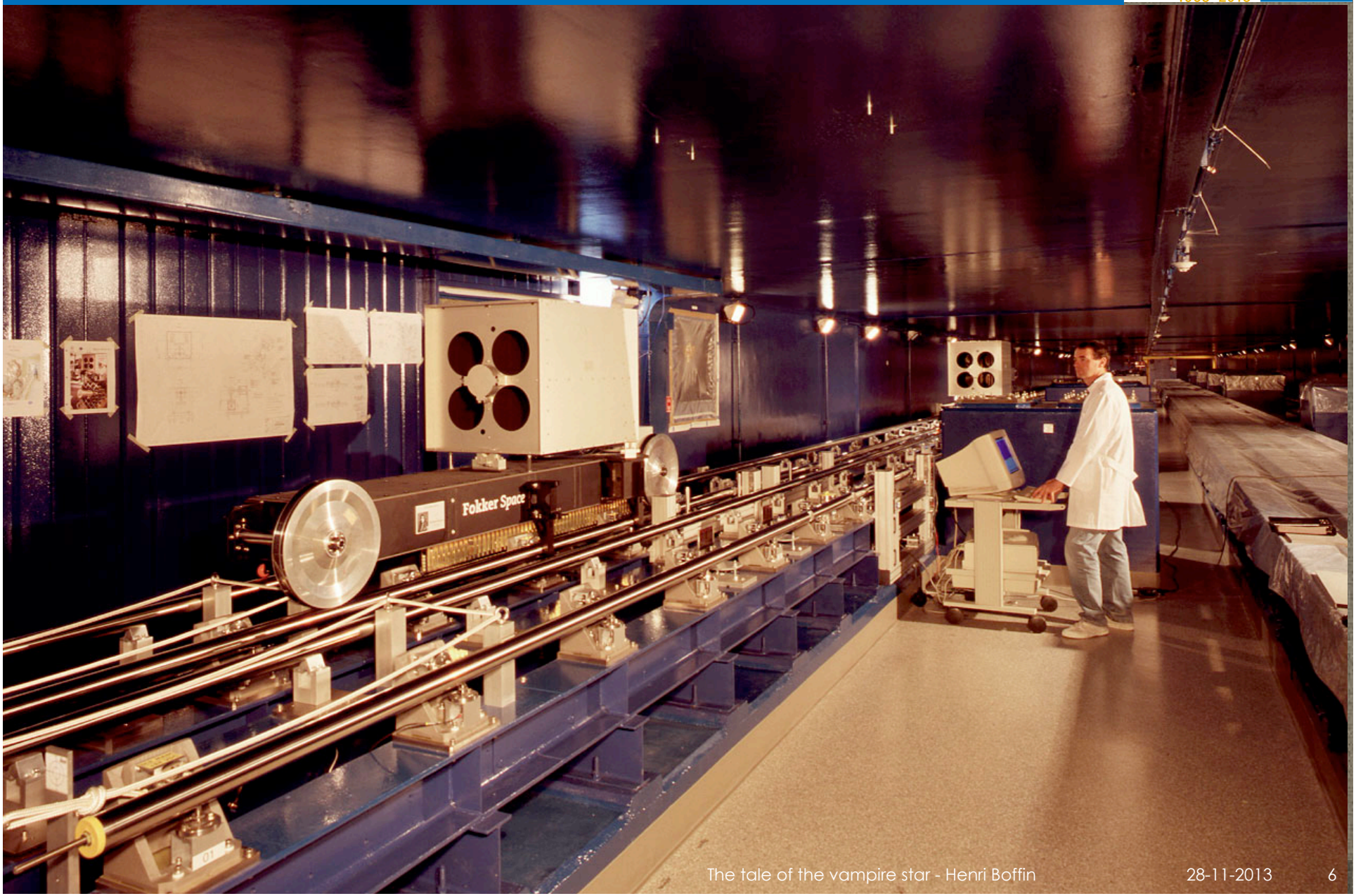
With interferometry, we can



VLT was conceived from start for Interferometry – combining light from telescopes

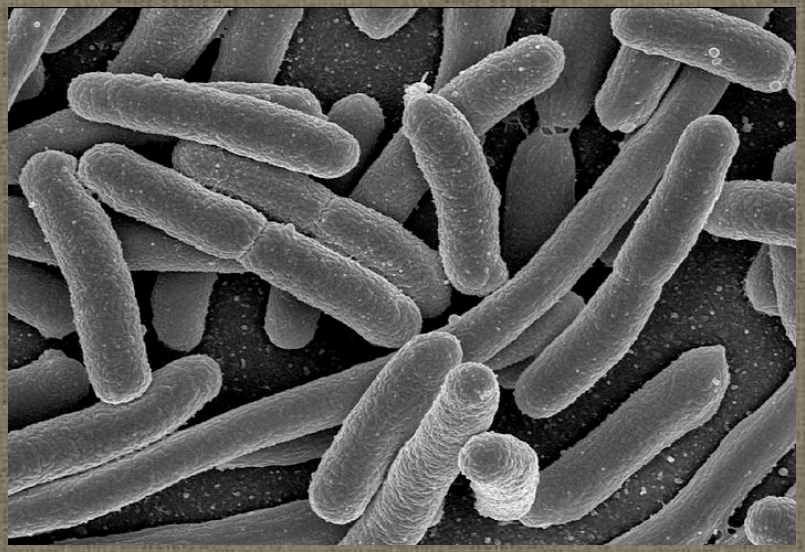


The interferometric laboratory



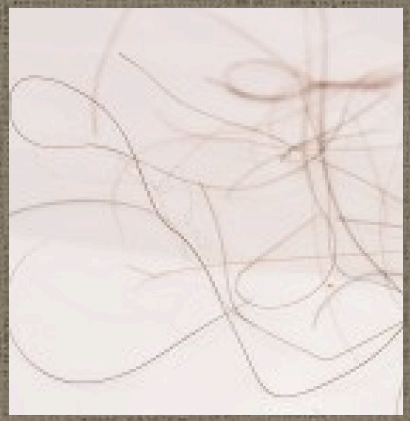
A technological challenge

the light paths must be kept equal to within 1/1000 mm – much less than the width of human hair – over distances of a few hundred metres.



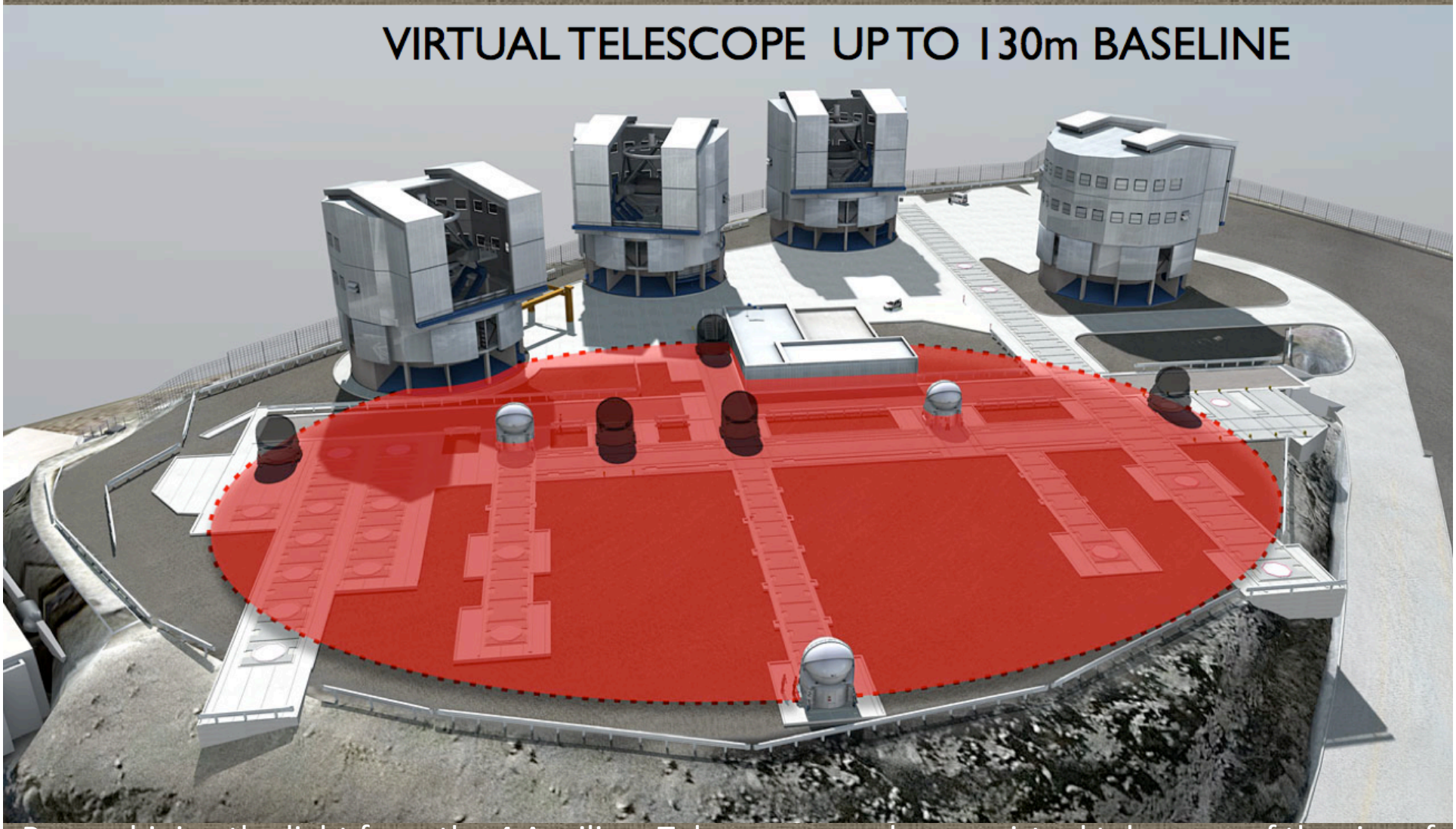
Credit: Rocky Mountain Laboratories, NIAID, NIH

An E. Coli cell is about a micron long, about the precision to which we need to keep the light paths equal!



Auxiliary Telescopes

Four 1.8-m telescopes
Built by the Belgian company AMOS
Can move on rails



By combining the light from the 4 Auxiliary Telescopes, we have a virtual telescope of the size of the largest separation between the telescopes

130-m Virtual Telescope

The VLTI provides a spatial resolution of 1 milliarcsecond. This is equivalent to picking out the head of a screw at a distance of 1200 km.

Paranal

1200 KM

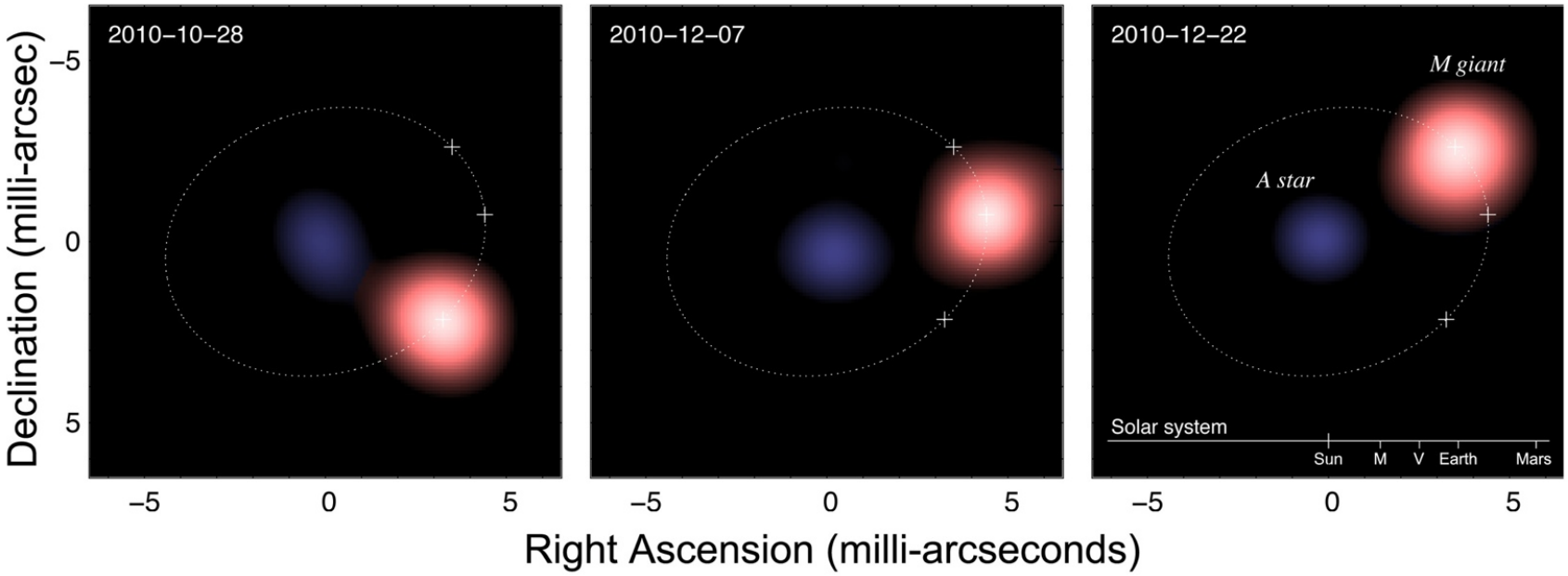
Santiago

An incredible technological achievement!

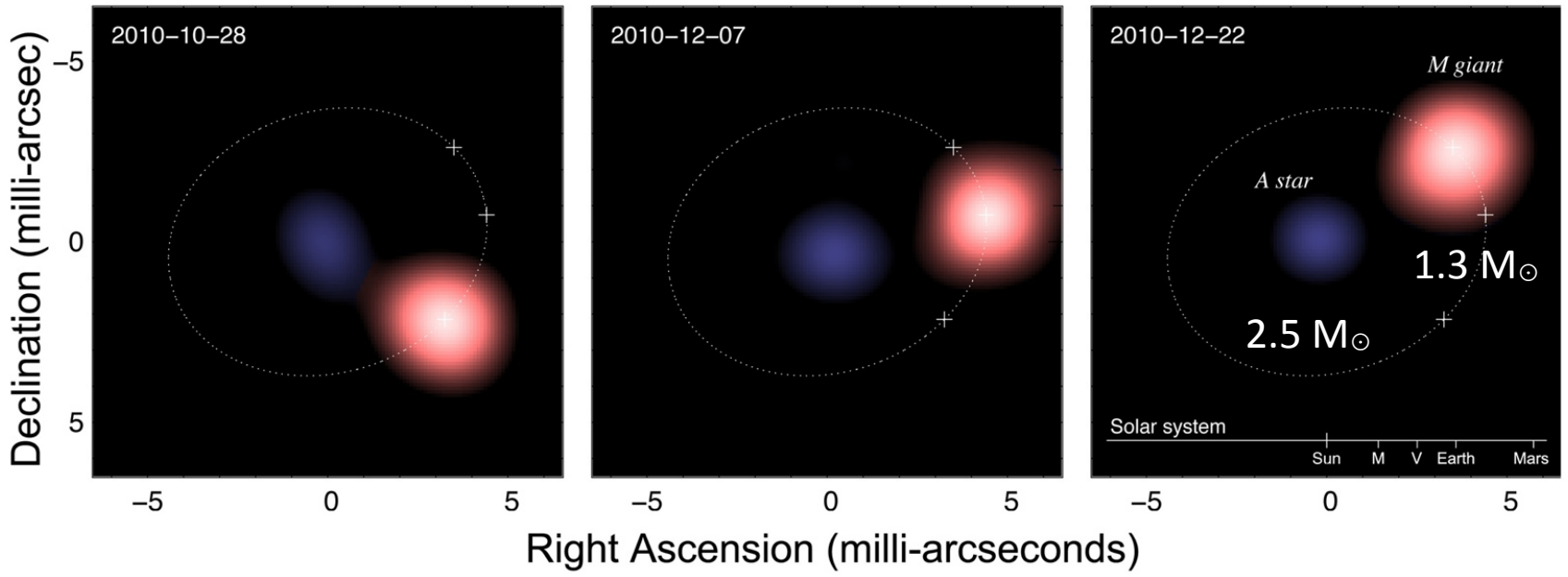
17 Leporis
A star visible with the naked eye

Using Interferometry: It is double!

Observing the system at different epochs, we can see the stars move around each other, similarly to the Earth moving around the Sun

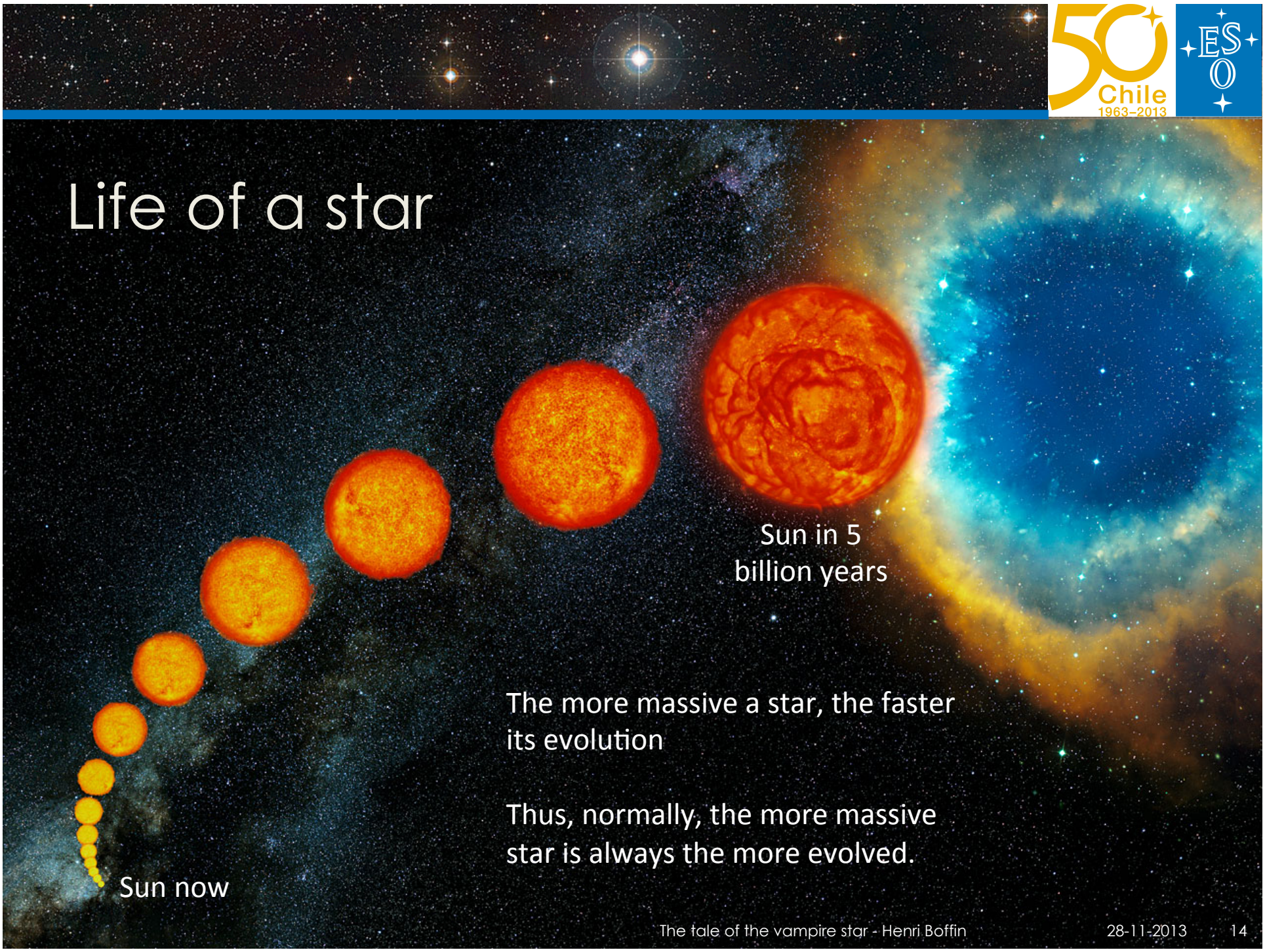


The VLTI allowed astronomers to measure the sizes of the stars and compute the orbit, from which one can determine the masses of the stars.



The system contains a red giant with 1.3 times the mass of the Sun and a blue star about twice as massive. The two stars orbit around each other in 260 days.

Life of a star



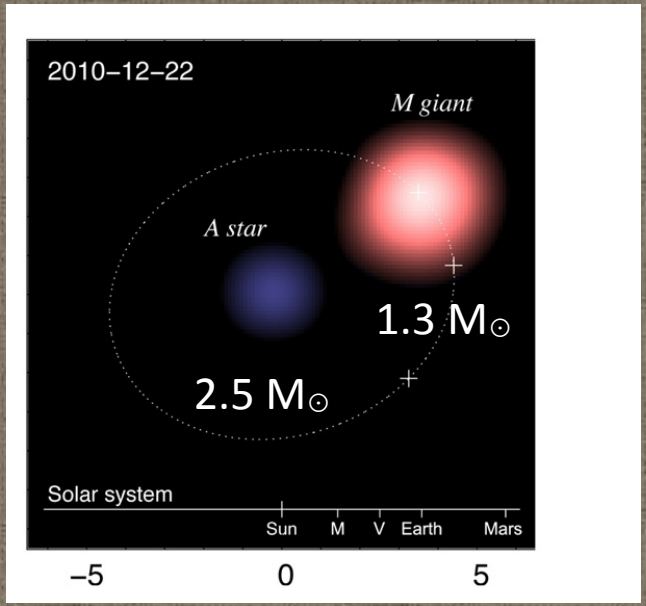
The more massive a star, the faster its evolution

Thus, normally, the more massive star is always the more evolved.

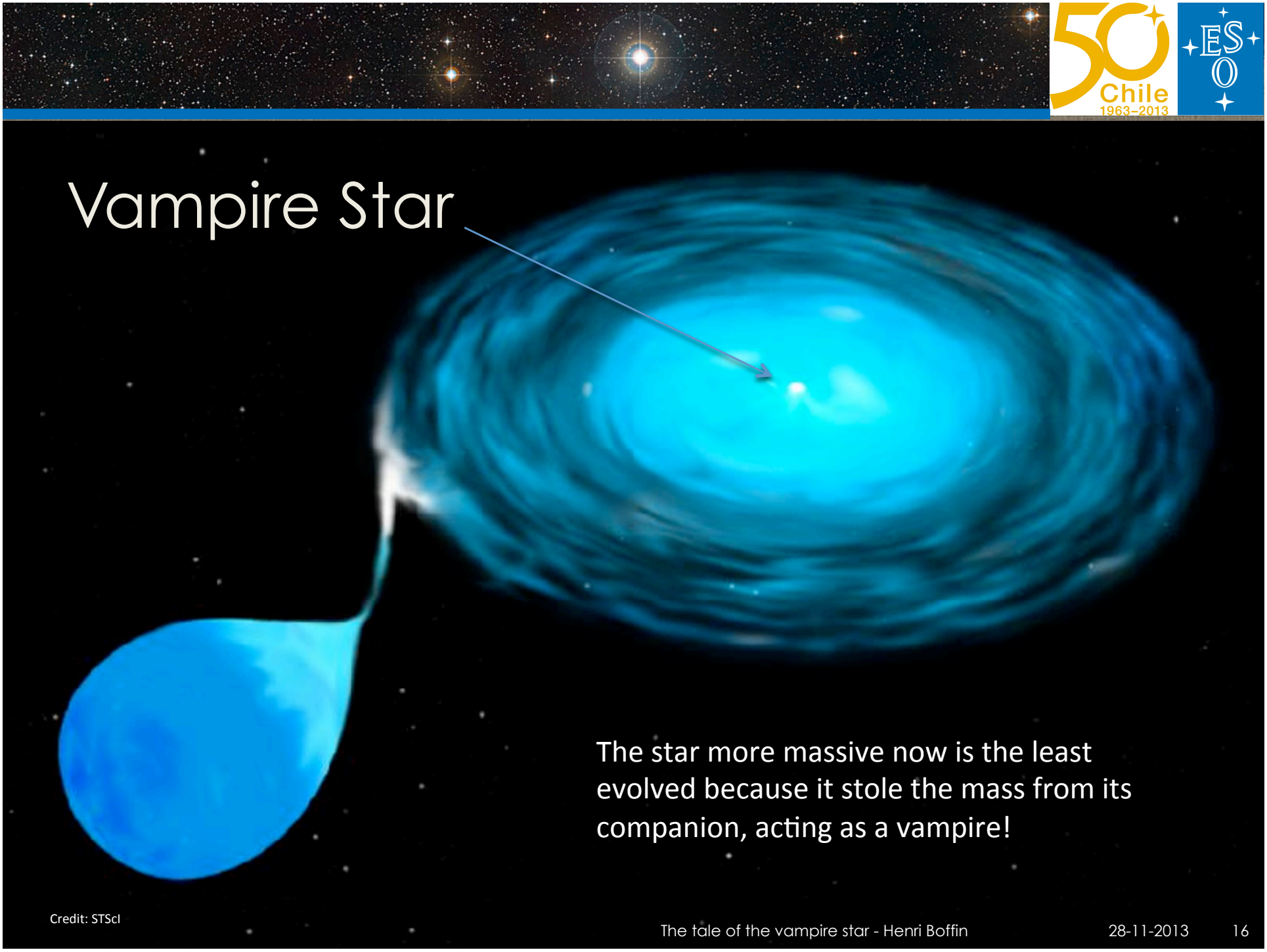
A mass reversal

Here, however, contrarily to what astronomers expect, the red giant (i.e. the most evolved star) is the less massive star!

Something dramatic happened!

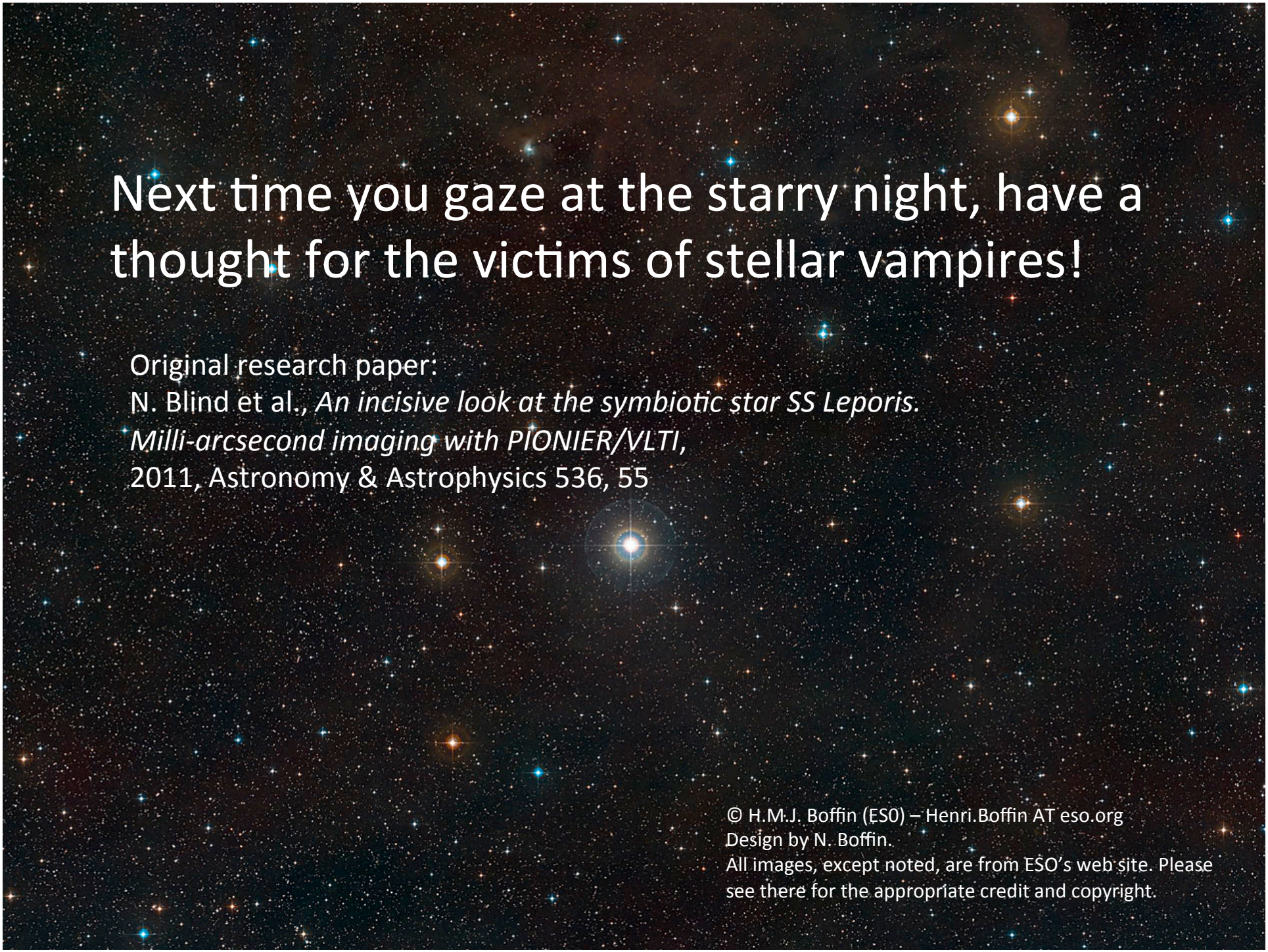


Vampire Star



The star more massive now is the least evolved because it stole the mass from its companion, acting as a vampire!

Credit: STScI



Next time you gaze at the starry night, have a thought for the victims of stellar vampires!

Original research paper:
N. Blind et al., *An incisive look at the symbiotic star SS Leporis. Milli-arcsecond imaging with PIONIER/VLTI*,
2011, Astronomy & Astrophysics 536, 55

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